

Gaurang Solanki

+1 (816) 838-8936 — gorangsolanki1@gmail.com — Portfolio — LinkedIn — GitHub

SUMMARY

Data Scientist with 3+ years of experience solving operational problems through applied AI across production environments. Bridges Deep Learning, computer vision, LLMs, and workflow automation to eliminate inefficiencies and deliver systems that create measurable business impact.

EXPERIENCE

Data Scientist Envision Building Products LLC
June 2024 – Present Lamar, MO

- Designed and optimized a YOLO-based real-time defect detection system for manufacturing quality control, reducing scrap rate by **17%** and overhead labor costs by **33%**.
- Owned the full computer vision pipeline, from data collection, image labeling, and augmentation to model training, evaluation, and production deployment across varied environments.
- Engineered automated data preprocessing and annotation workflows, maintaining high-quality datasets across thousands of production images for model retraining and validation.
- Built a Flask-based web application to automate drawing ETL workflows, improving turnaround time by **24%** and reducing drafting errors by **12%**.
- Architected and deployed a large-scale **RAG system** using Azure AI Search, centralizing thousands of documents across departments and enabling instant, context-aware access to production knowledge for the entire workforce.
- Developed and deployed custom **MCP servers** integrated with internal production systems, enabling workers to query and interpret live manufacturing data through natural language, significantly reducing response time and operational friction.
- Developed and deployed robotic process automations (RPAs) that eliminated **20+** manual work hours weekly across daily operations.

Graduate Assistant – Data Analyst Advising Department, University of Missouri–Kansas City
June 2023 – May 2024 Kansas City, MO

- Built Power BI dashboards tracking key performance metrics, increasing decision-making visibility for advisors and leadership.
- Consolidated data from multiple academic systems and automated weekly reporting, reducing manual effort and improving accuracy.

Jr. Machine Learning Engineer Omdena
May 2021 – July 2021 Remote

- Contributed to an end-to-end object detection and recommendation platform using **MobileNet SSD**, building a lightweight, production-ready dense detection system from scratch.
- Managed image annotation and quality verification pipelines, ensuring dataset integrity for training and validation of CV models.
- Improved deployment workflow efficiency, reducing operational overhead and enabling scalable model serving.

TECHNICAL SKILLS

ML & AI: Deep Learning, LLMs, RAG, Agentic AI, MCP, NLP, GNN, RNN, Machine Learning
Computer Vision: Object Detection, Image Classification, Image Segmentation, Visual Inspection, CNNs, Data Augmentation, OpenCV
Programming: Python, SQL, C/C++, JavaScript, HTML/CSS, R, MATLAB
Tools & Platforms: PyTorch, TensorFlow, Azure Services, DevOps, MLOps, Docker, Flask, FastAPI, Git, Spark, NoSQL, Power BI, Power Automate

EDUCATION

University of Missouri–Kansas City Aug 2022 – May 2024
Master of Science in Data Science — GPA: 3.83/4.0

Award: Discipline Distinction for academic excellence.
Research: Applied Graph Neural Networks and knowledge graphs for ontology-based image classification and visual summarization.
Projects: Facial Recognition & Validation System, Real-time Sentiment Analysis, ML-Based Diabetes Management Platform, Animal Migration due to Climate Change Analysis.

PUBLICATIONS & CERTIFICATIONS

- Publication:** *A Learning-Based Classification Model for Vegetables Recognition and Recipes Recommendation*, International Journal of Engineering Applied Science and Technology, Vol. 7(4), 2022.
- Certifications:** Microsoft Certified – Azure Fundamentals (2025), TensorFlow Developer (2021), Machine Learning – Stanford University (2022).
- Author of technical blogs on computer vision, deep learning, and ML systems — covering architecture design, training strategies, and deployment.